IRON WITHIN RESEARCH

BPC-157 / TB500 — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.

PAGE	STRENGTH	BATCH / LOT
2–3	10mg, 24mg, 31mg, 55mg	10/9/2026 · Lot IWR-1010-BPTB2
4–5	10mg, 17mg, 23mg, 40mg	9/13/2026 · Lot IWR-090926BPTB
6–14	3mg, 5mg, 10mg, 20mg, 22mg, 51mg, 52mg, 70mg, 82mg, 96mg, 99mg	8/28/2026
15–19	5mg, 6mg, 10mg, 15mg	02/22/2026

Page numbers include this index page. Testing is per production batch; one batch can cover several vial sizes.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0300

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
BPC-157 / TB-500 — 10 mg / 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-1010-BPTB2	99.62%	20.55 mg	Confirmed

COA #:	SBA-26-0300	Method:	Full QC Panel
Lot Number:	IWR-1010-BPTB2	Analysis Date:	2026-10-09
Labeled Content:	10 mg / 10 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	BPC-157 + TB-500	All confirmed
Purity (HPLC, aggregate)	>= 98.0%	99.62%
Blend Composition	Report only	Quantified — see composition
Content (per vial)	Report only	20.55 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.02 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth



SAMPLE AS RECEIVED

Test	Specification	Result
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

BLEND COMPOSITION

Mass basis, per-component reference standards

COMPONENT	AMOUNT	SHARE OF TOTAL	IDENTITY
BPC-157	**10.24 mg**	49.83%	Confirmed
TB-500	**10.31 mg**	50.17%	Confirmed

Each component is quantified on a mass basis against its own reference standard. Share of total will not match a component's area-percent in the chromatogram — different chromophores respond differently at the monitoring wavelength. Amounts are reported as measured; this certificate does not assess them against a declared formulation.



Kasey Malganius
Lab Director
2026-10-09

COA #: **SBA-26-0300**
Access Code: **42QYNZ8R**
Analyst: **A. Joseph**
Verify: summitbioanalytics.com/verify
Issued: 2026-10-09

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0187

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
BPC-157 / TB-500 — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-090926BPTB	99.91%	10.40 mg	Confirmed

COA #:	SBA-26-0187	Method:	Full QC Panel
Lot Number:	IWR-090926BPTB	Analysis Date:	2026-09-13
Labeled Content:	10 mg	Date Received:	2026-09-07
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Clear #A4A2A7

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	BPC-157 + TB-500	All confirmed
Purity (HPLC, aggregate)	>= 98.0%	99.91%
Blend Composition	Report only	Quantified — see composition
Content (per vial)	Report only	10.40 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.02 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

BLEND COMPOSITION

Mass basis, per-component reference standards

COMPONENT	AMOUNT	SHARE OF TOTAL	IDENTITY
BPC-157	**5.23 mg**	50.29%	Confirmed
TB-500	**5.17 mg**	49.71%	Confirmed

Each component is quantified on a mass basis against its own reference standard. Share of total will not match a component's area-percent in the chromatogram — different chromophores respond differently at the monitoring wavelength. Amounts are reported as measured; this certificate does not assess them against a declared formulation.



Kasey Malganius
Lab Director
2026-09-13

COA #:	**SBA-26-0187**
Access Code:	**TE6XF7VS**
Analyst:	**R. Jeffereson**
Verify:	summitbioanalytics.com/verify
Issued:	2026-09-13

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: IWR-8833606-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	BPC-157
Lot:	IWR-8833606-P
Appearance:	White Lyophilized Powder

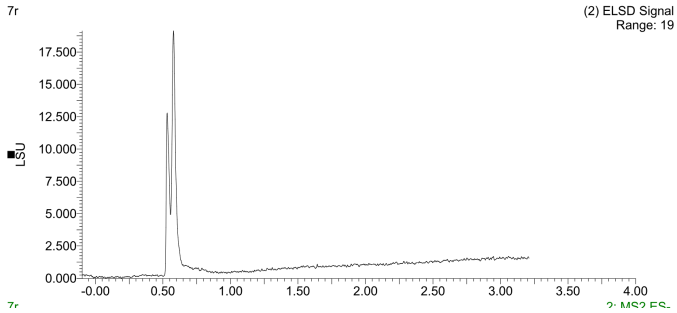
CAS:	137525-51-0
Formula:	$C_{62}H_{98}N_{16}O_{22}$
Mol Weight:	~1419.5 g/mol

Pubchem CID: 108101

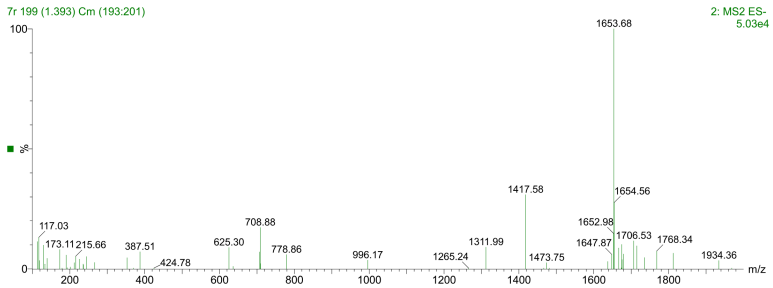
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	BPC-157	BPC-157	
Quantity:	5mg	5.03mg	
Purity:	≥98%	99.29%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

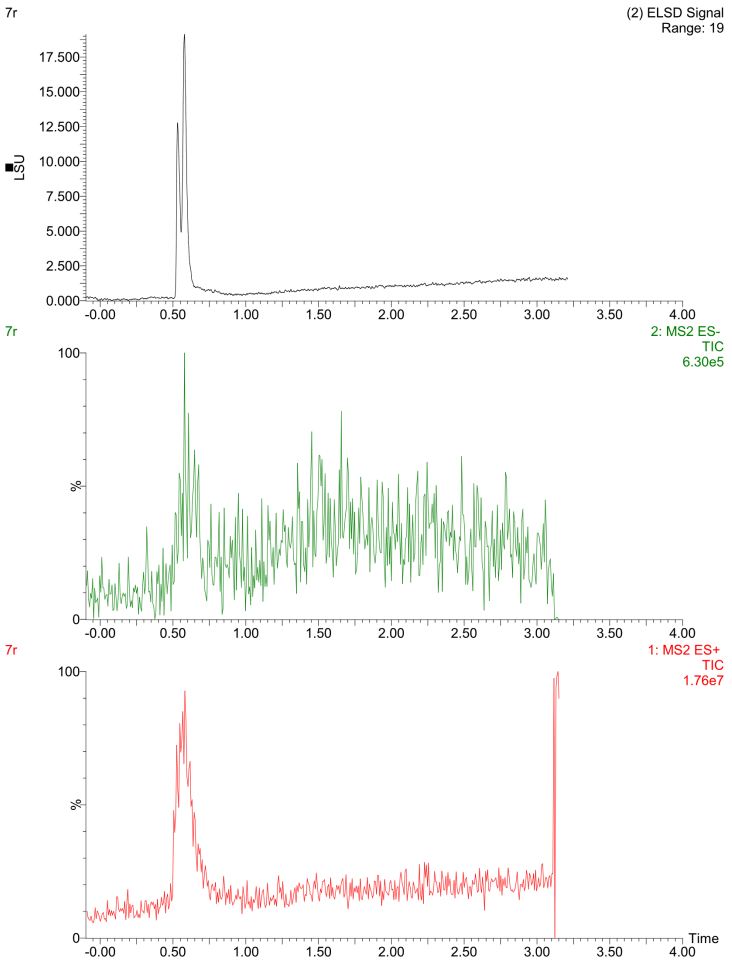
Certificate of Analysis

Lot Number: IWR-8833606-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

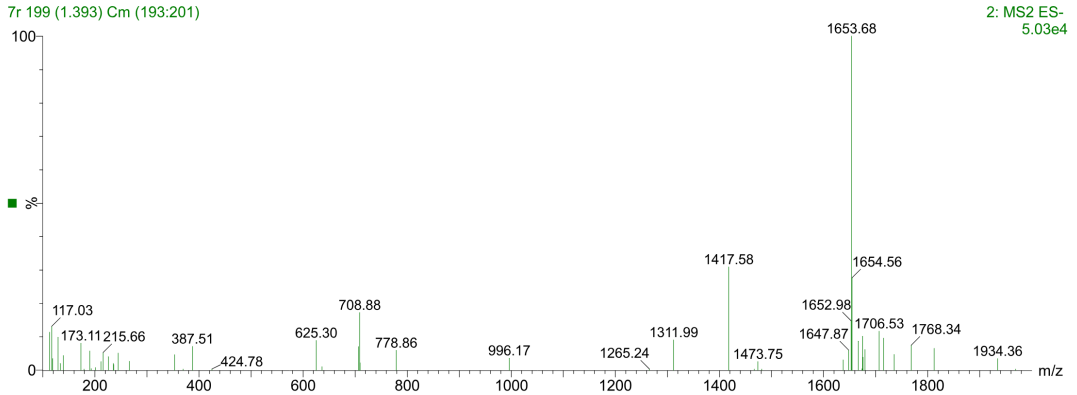
Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

BPC-157 (5mg) • Pubchem CID: 108101

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-8833606-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	TB-500
Lot:	IWR-8833606-P
Appearance:	White Lyophilized Powder

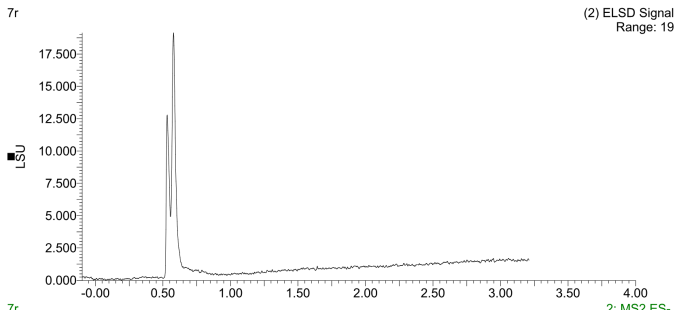
CAS:	77591-33-4
Formula:	$C_{212}H_{350}N_{56}O_{78}S$
Mol Weight:	~4963 g/mol

Pubchem CID: 16132341

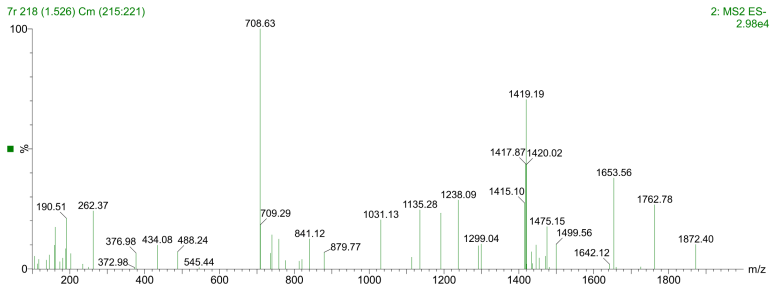
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	TB-500	TB-500	
Quantity:	5mg	5.03mg	
Purity:	≥98%	99.32%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

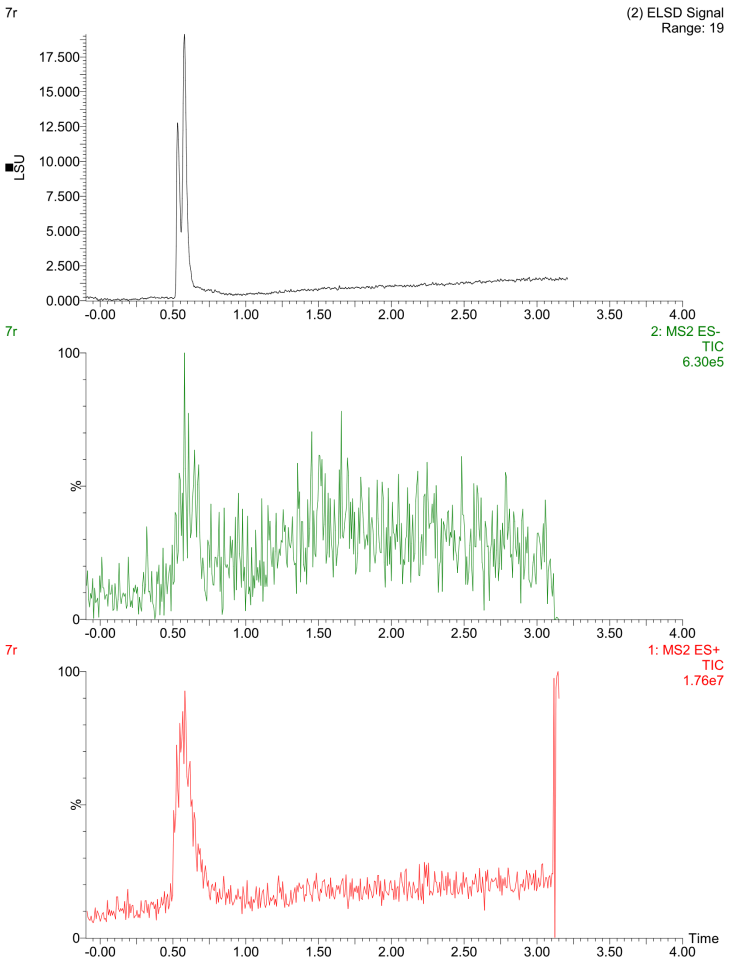
Certificate of Analysis

Lot Number: IWR-8833606-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

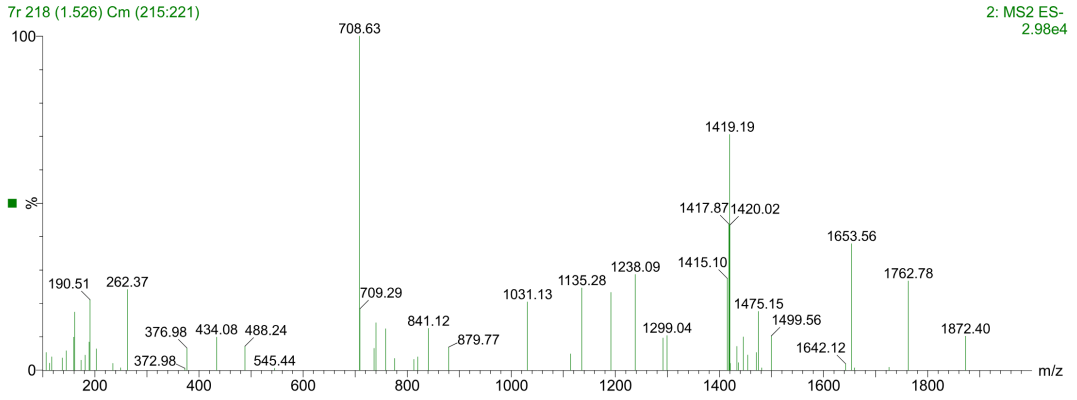
Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

TB-500 (5mg) • Pubchem CID: 16132341

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-8833606-E
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	BPC-157 + TB-500
Lot:	IWR-8833606-E
Appearance:	-

CAS:	BPC-157, TB-500
Formula:	N/A
Mol Weight:	N/A

Pubchem CID: N/A

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	BPC-157 + TB-500	-	
Endotoxin:	-	< 0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

Proudly Owned and Operated in the USA

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0029

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
TB-500 / BPC-157 — 20 mg

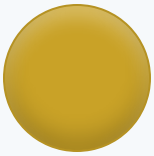
PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-BPCTB20	99.76%	22.51 mg	Confirmed

COA #:	SBA-26-0029	Method:	Full QC Panel
Lot Number:	IWR-2607-BPCTB20	Analysis Date:	2026-08-02
Labeled Content:	20 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free



VIAL CAP
Gold
#C9A227

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	TB-500 + BPC-157	All confirmed
Purity (HPLC, aggregate)	>= 98.0%	99.76%
Blend Composition	Report only	Quantified — see composition

Test	Specification	Result
Content (gross)	Report only	**22.51 mg**
Fentanyl Screen (LC-MS/MS)	Not detected	**Not detected**
Endotoxins (LAL)	Report Result	**0.07 EU/mL**
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

BLEND COMPOSITION

Mass basis, per-component reference standards

COMPONENT	AMOUNT	SHARE OF TOTAL	IDENTITY
TB-500	**11.52 mg**	51.18%	Confirmed
BPC-157	**10.99 mg**	48.82%	Confirmed

Each component is quantified on a mass basis against its own reference standard. Share of total will not match a component's area-percent in the chromatogram — different chromophores respond differently at the monitoring wavelength. Amounts are reported as measured; this certificate does not assess them against a declared formulation.

Kasey Malganius

Kasey Malganius
Lab Director
2026-08-02

COA #: **SBA-26-0029**
Access Code: **EHBAG2GL**
Analyst: **R. Whitfield**
Verify: summitbioanalytics.com/verify
Issued: 2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0030

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Ac-SDKP / TB-500 / BPC-157 — 10 mg

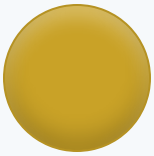
PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-BPCTB10	99.31%	11.48 mg	Confirmed

COA #:	SBA-26-0030	Method:	Full QC Panel
Lot Number:	IWR-2607-BPCTB10	Analysis Date:	2026-08-02
Labeled Content:	10 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free



VIAL CAP
Gold
#C9A227

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Ac-SDKP + TB-500 + BPC-157	All confirmed
Purity (HPLC, aggregate)	>= 98.0%	99.31%

Test	Specification	Result
Blend Composition	Report only	**Quantified — see composition**
Content (gross)	Report only	**11.48 mg**
Fentanyl Screen (LC-MS/MS)	Not detected	**Not detected**
Endotoxins (LAL)	Report Result	**0.02 EU/mL**
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

BLEND COMPOSITION

Mass basis, per-component reference standards

COMPONENT	AMOUNT	SHARE OF TOTAL	IDENTITY
Ac-SDKP	**3.82 mg**	33.28%	Confirmed
TB-500	**1.96 mg**	17.07%	Confirmed
BPC-157	**5.70 mg**	49.65%	Confirmed

Each component is quantified on a mass basis against its own reference standard. Share of total will not match a component's area-percent in the chromatogram — different chromophores respond differently at the monitoring wavelength. Amounts are reported as measured; this certificate does not assess them against a declared formulation.



Kasey Malganius
Lab Director
2026-08-02

COA #: **SBA-26-0030**
Access Code: **YQ26KY8L**
Analyst: **R. Whitfield**
Verify: summitbioanalytics.com/verify
Issued: 2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: IWR-7454331-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Compound:	BPC-157
Lot:	IWR-7454331-P
Appearance:	White Lyophilized Powder

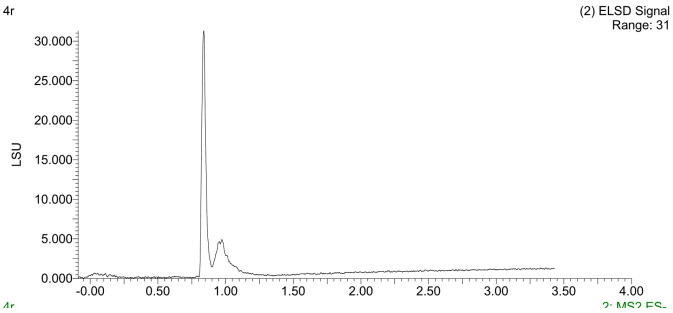
CAS:	137525-51-0
Formula:	$C_{62}H_{98}N_{16}O_{22}$
Mol Weight:	~1419.5 g/mol

Pubchem CID: 108101

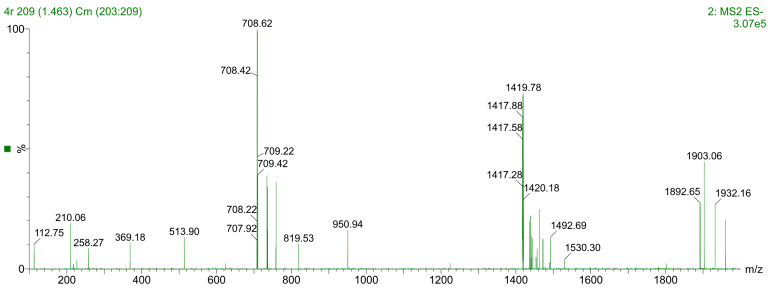
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	BPC-157	BPC-157	
Quantity:	5mg	9.90	
Purity:	≥98%	99.36	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

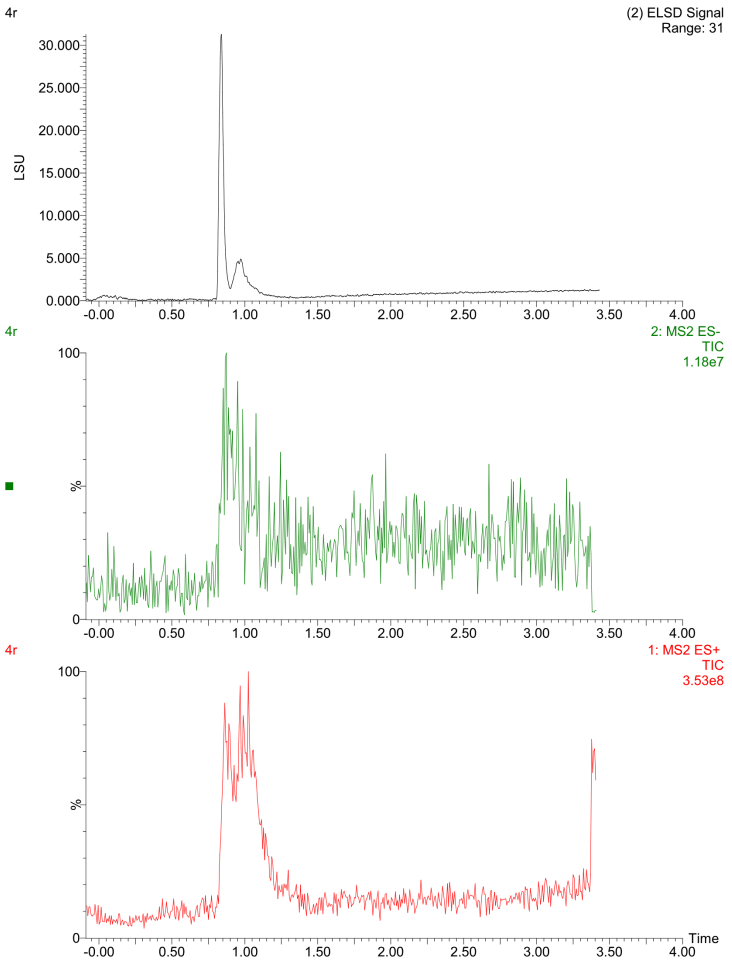
Certificate of Analysis

Lot Number: IWR-7454331-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

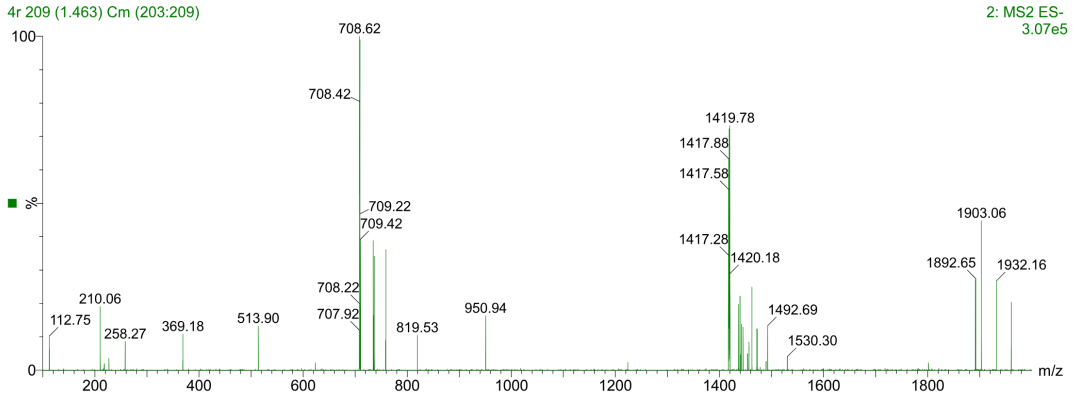
Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

BPC-157 (5mg) • Pubchem CID: 108101

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-7454331-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Compound:	TB-500
Lot:	IWR-7454331-P
Appearance:	White Lyophilized Powder

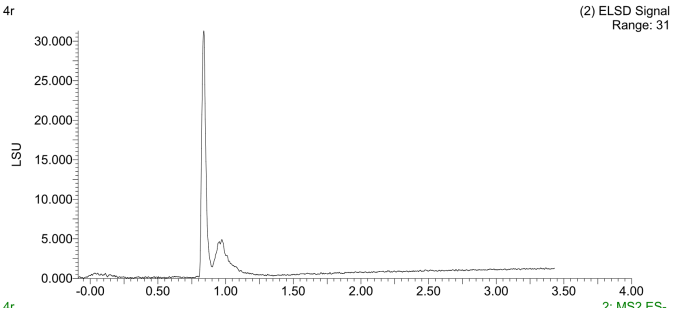
CAS:	77591-33-4
Formula:	$C_{212}H_{350}N_{56}O_{78}S$
Mol Weight:	~4963 g/mol

Pubchem CID: 16132341

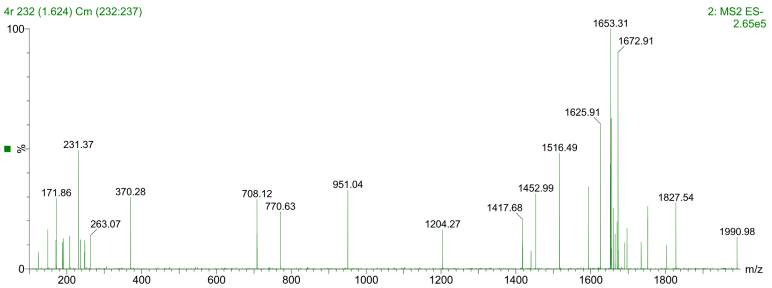
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	TB-500	TB-500	
Quantity:	5mg	9.90	
Purity:	≥98%	99.19%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

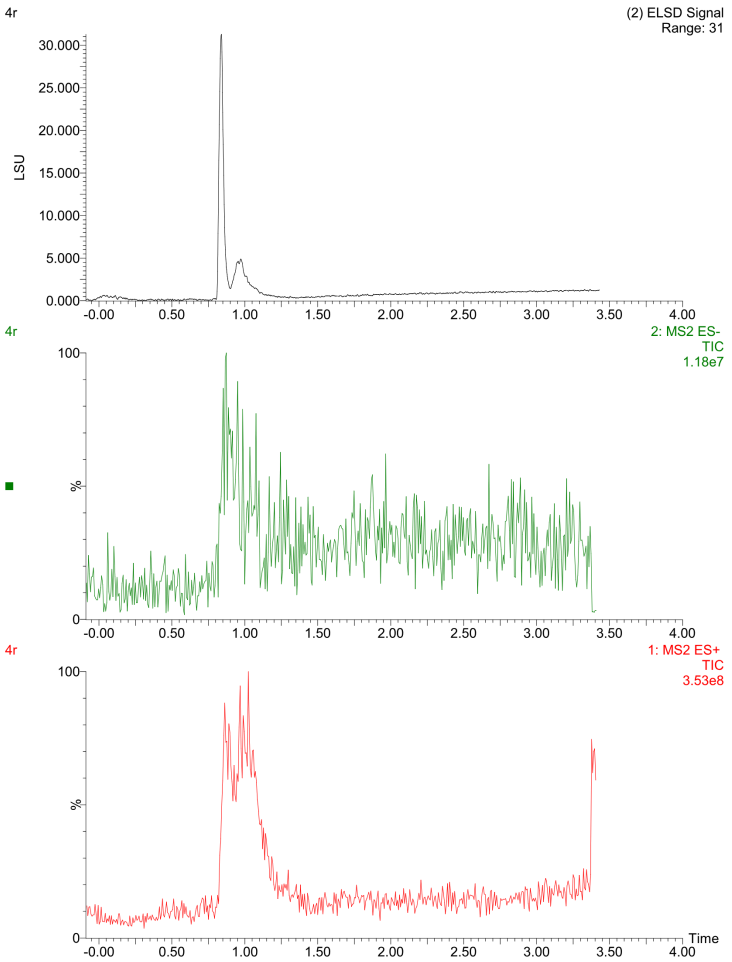
Certificate of Analysis

Lot Number: IWR-7454331-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

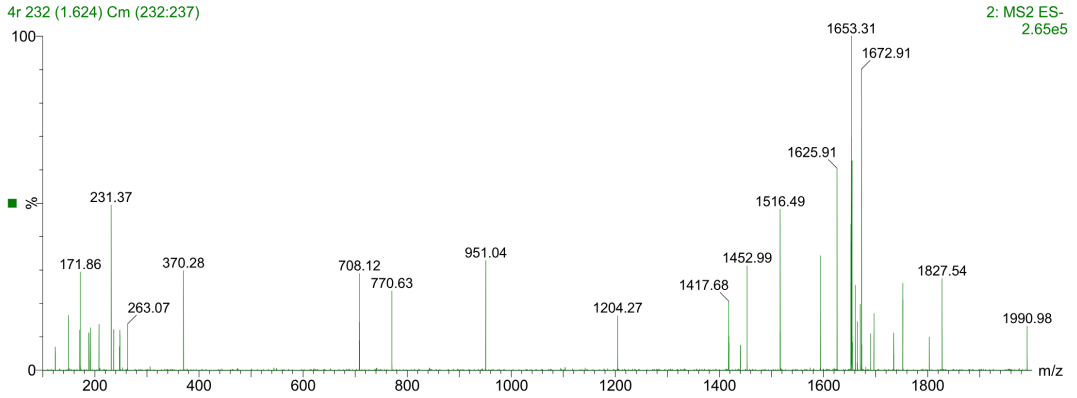
Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

TB-500 (5mg) • Pubchem CID: 16132341

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Certificate of Analysis

FREEDOM DIAGNOSTICS

Proudly Owned and Operated in the USA

Searchable via: FreedomDiagnosticsTesting.com

Accession Number	2602200198
Client	Iron Within Nutrition LLC
Search Code	Iron2602200198

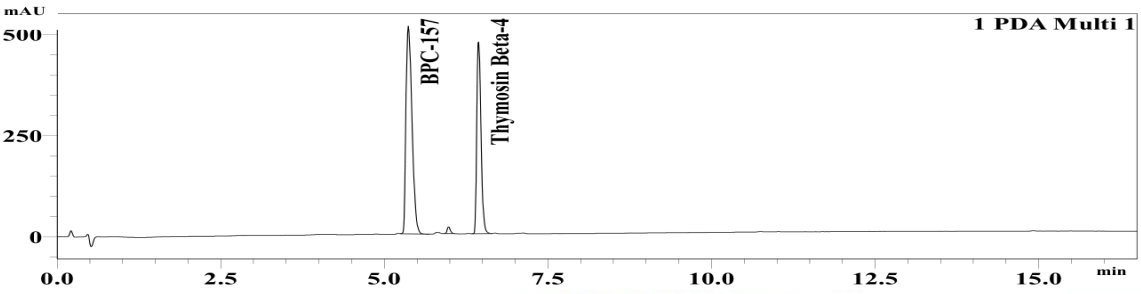
Received Date:	02/20/2026
Reported Date:	2/22/2026

Product	BPC 10mg + TB 10mg
Net Peptide Content	BPC-157 – 11.15 mg Thymosin Beta-4 – 11.06 mg
Identity	BPC-157/Thymosin Beta-4

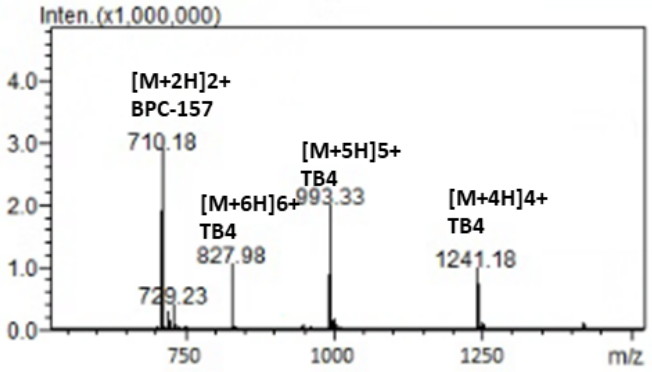
Lot	Purple Cap
Purity	99.009%
Appearance	White Lyophilized Powder

All Chemical Analysis was performed by HPLC with UV Detection Coupled with Mass Spectrometry

Mass Identification	Result
BPC-157/Thymosin Beta-4	**99.009%**







Stephen Schmidt

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Principal Chemist

COA: 2602200198

The peptide purity analysis reported here was conducted using LCMS/MS under standard laboratory conditions. This analysis is intended for informational purposes only and is specific to the sample(s) provided. The peptides tested are intended for research use only and are not approved for human or veterinary use, diagnostic, therapeutic, or clinical applications. Results should be interpreted by qualified professionals within the scope of the intended research. The accuracy and reliability of the test may be influenced by sample integrity, handling, and other experimental variables.